

26020 Chemistry (Polytechnic Foundation) winter 2025/Kemi (Polyteknisk Grundlag) vinter 2025 non random

”.” bruges som decimalseparator på både dansk og engelsk.

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

Der er altid netop ét svar som er mere rigtigt end de andre

Studerende kan kun vælge ét svar per spørgsmål

Hvert rigtigt svar giver 1 point

Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

”.” is used as decimal separator in the both Danish and English text.

The following approach to scoring responses is implemented and is based on "One best answer"

There is always only one correct answer – a response that is more correct than the rest

Students are only able to select one answer per question

Every correct answer corresponds to 1 point

Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

DK: Betragt følgende: ^{180}Hf , ^{180}Ta , ^{180}W . For disse atomer gælder:

EN: Consider the following: ^{180}Hf , ^{180}Ta , ^{180}W . These atoms have:

Choose one answer

- ☐ Det samme antal elektroner. *The same number of electrons.*
- ☐ Det samme nukleontal (mass number). *The same mass number.*
- ☐ Det samme antal neutroner. *The same number of neutrons.*
- ☐ Det samme antal protoner. *The same number of protons.*
- ☐ Det samme antal protoner og det samme antal neutroner. *The same number of protons and the same number of neutrons.*

DK: Hvor mange elektroner har selen i sin yderste skal - valensskallen?

EN: *How many electrons does selenium possess in its outermost shell - the valence shell?*

Choose one answer

- ☐ 1
- ☐ 6
- ☐ 4
- ☐ 2
- ☐ 7
- ☐ 3
- ☐ 5

DK: Kogepunkterne for hydrogenfluorid (HF), vand (H₂O), ammoniak (NH₃), og metan (CH₄) er henholdsvis 20°C, 100°C, -33°C og -162°C. Hvad er forklaringen på methans meget lave kogepunkt?

EN: *The boiling points of hydrogen fluoride (HF), water (H₂O), ammonia (NH₃), and methane (CH₄) are 20°C, 100°C, -33°C, and -162°C, respectively. What explains the very low boiling point of methane?*

Choose one answer

- ☐ Ingen hydrogenbinding. *No hydrogen bonding*
- ☐ Den molare masse. *The molar mass.*
- ☐ Ingen ion-binding. *No ionic bonding*
- ☐ Kovalent binding. *Covalent bonding.*
- ☐ Ingen van der Waals-vekselvirkninger. *No van der Waals interactions.*
- ☐ Ingen ion-dipol-vekselvirkninger. *No ion-dipole interactions.*

DK: Hvilken af følgende ioner har den største ionradius?

EN: Which of the following ions has the largest ionic radius?

Choose one answer

☐ Na^+

☐ Li^+

☐ K^+

☐ Mg^{2+}

☐ Ca^{2+}

☐ Be^{2+}

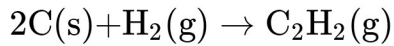
DK: Hvor mange mol ioner dannes ved fuldstændig opløsning af 1.5 mol strontiumchlorid i vand?

EN: *How many moles of ions are formed by complete dissolution of 1.5 mol of strontium chloride in water?*

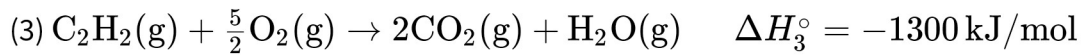
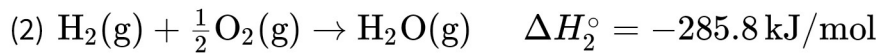
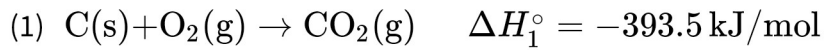
Choose one answer

- ☐ 1.0 mol
- ☐ 4.5 mol
- ☐ 2 mol
- ☐ 1.5 mol
- ☐ 3 mol
- ☐ 5 mol

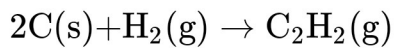
DK: Udregn ΔH° for reaktionen:



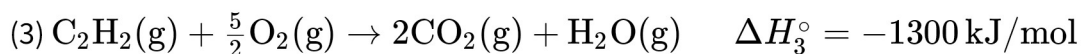
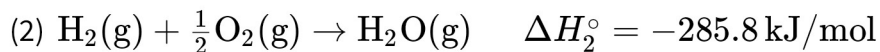
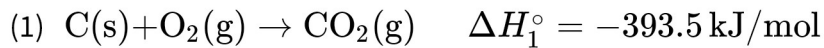
givet informationen:



EN: Calculate ΔH° for the reaction



given the information:



Choose one answer

- ☐ -620.7 kJ/mol
- ☐ -227.2 kJ/mol
- ☐ 227.2 kJ/mol
- ☐ 620.7 kJ/mol
- ☐ 1072.8 kJ/mol
- ☐ -1072.8 kJ/mol

DK: Atomet med elektronkonfigurationen $3s^23p^3$ i yderste skal vil være i:

EN: The atom with the electronic configuration $3s^23p^3$ in the outermost shell will be in:

Choose one answer

- ☐ 3. hovedgruppe, 3. periode. 3^{rd} main group, 3^{rd} period.
- ☐ 3. hovedgruppe, 5. periode. 3^{rd} main group, 5^{th} period.
- ☐ 5. hovedgruppe, 3. periode. 5^{th} main group, 3^{rd} period.
- ☐ 3. hovedgruppe, 6. periode. 3^{rd} main group, 6^{th} period.
- ☐ 6. hovedgruppe, 3. periode. 6^{th} main group, 3^{rd} period.

DK: Hvilket udsagn er korrekt om SO_2 (som har svovl som centralatom)?

EN: Which statement is correct for SO_2 (which has sulfur as a central atom)?

Choose one answer

- ☐ Molekylet er vinklet og polært. *The molecule is bent and polar.*
- ☐ Molekylet er vinklet og upolært. *The molecule is bent and non-polar.*
- ☐ Molekylet er lineært og polært. *The molecule is linear and polar.*
- ☐ Molekylet er lineært og upolært. *The molecule is linear and non-polar.*

DK: Hvad er den formelle ladning på nitrogenatomet i nitrit (NO_2^-) bestemt ud fra Lewis-strukturen?

EN: What is the formal charge on the nitrogen atom in nitrite (NO_2^-) as determined from the Lewis structure?

Choose one answer

- ☐ 0
- ☐ +2
- ☐ -2
- ☐ -1
- ☐ +1

DK: Hvad er den omtrentlige forventede H-O-O bindingsvinkel i H_2O_2 (hydrogenperoxid) baseret på Lewis strukturen?

EN: *What is the approximate expected H-O-O bond angle in H_2O_2 (hydrogen peroxide) based on the Lewis structure?*

Choose one answer

- ☐ ~109.5°
- ☐ ~120°
- ☐ ~90°
- ☐ ~180°
- ☐ ~54°

DK: Hvilket af følgende molekyler har en plan geometri?

EN: Which of the following molecules possesses a planar geometry?

Choose one answer

☐ PH_3

☐ CH_4

☐ C_2H_4

☐ PCl_3

☐ SF_6

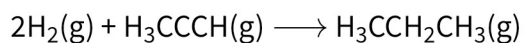
DK: I et horisontalt glaserør placeres i hver sin ende to små bægerglas med henholdsvis koncentreret vandig ammoniak (NH_3) og vandig hydrogenbromid (HBr). Glasrørets ender lukkes således at luften i røret er i ro. Hvad kan observeres efter noget tid?

EN: *In a horizontal glass tube, two small beakers are placed at each end containing concentrated aqueous ammonia (NH_3) and aqueous hydrogen bromide (HBr), respectively. The ends of the glass tube are sealed such that the air in the tube is at rest. What can be observed after some time?*

Choose one answer

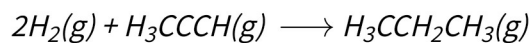
- ☐ Der dannes en hvid ring (som består af NH_4Br) midt i røret. *A white ring (which consists of NH_4Br) is formed in the middle of the tube.*
- ☐ Der dannes en hvid ring (som består af NH_4Br) i røret tættest på bægeret med hydrogenbromidopløsning. *A white ring is formed (which consists of NH_4Br) in the tube closest to the beaker with hydrogen bromide solution.*
- ☐ Der dannes en hvid sky (som består af NH_4Br) jævnt i hele røret. *A white cloud (which consists of NH_4Br) is formed uniformly in the tube.*
- ☐ Der dannes en hvid ring (som består af NH_4Br) i røret tættest på bægeret med ammoniakopløsning. *A white ring is formed (which consists of NH_4Br) in the tube closest to the beaker with ammonia solution.*
- ☐ Der sker ingenting. *Nothing is to be seen.*

DK: Estimer ΔH for følgende reaktion



givet bindingsenthalpierne for H-H (-436 kJ/mol), C-C (-350 kJ/mol), C-H (-410 kJ/mol) og $\text{C} \equiv \text{C}$ (-835 kJ/mol)

EN: Estimate ΔH for the following reaction



given the bond enthalpies for H-H (-436 kJ/mol), C-C (-350 kJ/mol), C-H (-410 kJ/mol) og $\text{C} \equiv \text{C}$ (-835 kJ/mol)

Choose one answer

- ☐ 589 kJ/mol
- ☐ -1103 kJ/mol
- ☐ -291 kJ/mol
- ☐ -2031 kJ/mol
- ☐ 202 kJ/mol

DK: Hvilken af følgende forbindelser har det laveste damptryk ved stuetemperatur?

EN: Which of the following possesses the lowest vapor pressure at room temperature?

Choose one answer

- ☐ Eddikesyre / *Acetic Acid*
- ☐ Methanol / *Methanol*
- ☐ Dichlormethan / *Dichloromethane*
- ☐ Chlormethan / *Chloromethane*
- ☐ Methan / *Methane*

DK: Der fremstilles 2 opløsninger:

1: 1 g NaCl opløses i 500 g vand.

2: 1 g CaCl₂ opløses i 500 g vand.

Frysepunkterne for de to opløsninger, $T_{m,1}$ og $T_{m,2}$, måles. Hvilken af nedenstående udsagn er korrekt?

EN:

Two solutions are prepared:

1: 1 g NaCl is dissolved in 500 g water.

2: 1 g CaCl₂ is dissolved in 500 g water.

The freezing points of the two solutions, $T_{m,1}$ and $T_{m,2}$, are measured. Which of the following statements is correct?

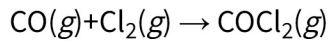
Information: $M_{\text{NaCl}} = 58.44 \text{ g/mol}$; $M_{\text{CaCl}_2} = 110.98 \text{ g/mol}$. van't Hoff

Factors $i_{\text{NaCl}}^* = 1.8$; $i_{\text{CaCl}_2}^* = 2.6$

Choose one answer

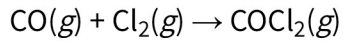
- ☐ $T_{m,1} < T_{m,2} < 0^\circ\text{C}$
- ☐ $T_{m,2} < T_{m,1} < 0^\circ\text{C}$
- ☐ $T_{m,1} < 0^\circ\text{C}$; $T_{m,2} > 0^\circ\text{C}$
- ☐ $T_{m,1} = T_{m,2}$
- ☐ $T_{m,1} > 0^\circ\text{C}$; $T_{m,2} > 0^\circ\text{C}$
- ☐ $T_{m,1} > 0^\circ\text{C}$; $T_{m,2} < 0^\circ\text{C}$

DK: Bestem reaktionsordenen for kulmonoxid (CO) og chlor (Cl₂) i hastighedsudtrykket, $hastighed = k p(\text{CO})^x p(\text{Cl}_2)^y$ hvor k er en konstant for reaktionen



hvor følgende data er blevet bestemt (se under den engelske tekst):

EN: Determine the reaction order for carbon monoxide (CO) and chlorine (Cl₂) in the rate expression $rate = k p(\text{CO})^x p(\text{Cl}_2)^y$ where k is a constant for the reaction



where the following data have been determined:

experiment number	[CO] (atm)	[Cl ₂] (atm)	rate (atm/s)
1	0.5	2	0.03
2	0.75	2	0.045
3	0.5	1	0.015

Choose one answer

- ☐ x=2; y=2
- ☐ x=1; y=1
- ☐ x=1; y=2
- ☐ x=1.5; y=1.5
- ☐ x=2; y=1

DK: Hvilket udsagn er ikke korrekt om en katalysators påvirkning på en kemisk reaktion?

EN: Which statement is not correct for a catalyst's influence on a chemical reaction?

Choose one answer

- ☐ Katalysatorer ændrer ikke ligevægtskonstanten. *Catalysts do not change the equilibrium constant.*
- ☐ Katalysatorer virker kun ved høj temperatur. *Catalysts only work at high temperatures.*
- ☐ Katalysatorer forøger reaktionshastigheden. *Catalysts speed up chemical reactions.*
- ☐ Katalysatorer kan være heterogene eller homogene. *Catalysts can be heterogeneous or homogeneous.*
- ☐ Katalysatorer forbruges ikke i reaktionen. *Catalysts are not consumed in the reaction.*

DK: 10 mg AgBr og 100 mg CuBr tilsættes til 1 L opløsning af 1.0×10^{-6} mol NaBr i vand. Hvad sker der med AgBr og CuBr når det oplyses at de relevante opløslighedsprodukter er:

$$K_{sp}(\text{AgBr}) = 3.3 \times 10^{-13} \text{ og } K_{sp}(\text{CuBr}) = 5.3 \times 10^{-9}$$

EN: 10 mg AgBr and 100mg CuBr are added to 1 L solution of 1.0×10^{-6} mol NaBr in water. What happens to the AgBr and CuBr given the relevant solubility products:

$$K_{sp}(\text{AgBr}) = 3.3 \times 10^{-13} \text{ and } K_{sp}(\text{CuBr}) = 5.3 \times 10^{-9}$$

Choose one answer

- ☐ Både AgBr og CuBr kan opløses. *Both AgBr and CuBr can dissolve*
- ☐ AgBr opløses fulstændigt mens CuBr ikke kan opløses. *AgBr dissolves completely while CuBr cannot dissolve.*
- ☐ Hverken AgBr eller CuBr kan opløses. *Neither AgBr nor CuBr can dissolve.*
- ☐ Det kan ikke afgøres. *It cannot be decided*
- ☐ CuBr opløses fulstændigt mens AgBr ikke kan opløses. *CuBr dissolves completely while AgBr cannot dissolve.*

DK: Hvor meget bariumsulfat (BaSO_4) kan opløses i 100 mL vand.

EN: How much barium sulfate (BaSO_4) can be dissolved in 100 mL of water.

$K_{sp}(\text{BaSO}_4) = 1.08 \times 10^{-10}$; $M_{\text{Ba}} = 137.33 \text{ g/mol}$; $M_{\text{O}} = 16.00 \text{ g/mol}$; $M_{\text{S}} = 32.06 \text{ g/mol}$;

Choose one answer

- ☐ 0.24 mg
- ☐ 2.4 mg
- ☐ 23.3 g
- ☐ 108 ng
- ☐ 11 ng

DK: Hvad er pH i en 0.10 M vandig opløsning af ammoniak (NH_3 , $pK_b = 4.75$).

EN: What is the pH of a 0.10 M aqueous solution of ammonia (NH_3 , $pK_b = 4.75$).

Choose one answer

- ☐ 13.0
- ☐ 11.1
- ☐ 9.25
- ☐ 8.75
- ☐ 7.0

DK: For hvilken af følgende reaktioner vil et fald i trykket favorisere dannelsen af produkterne? (se reaktioner under engelsk tekst).

EN: For which of the following reactions would a decrease in the pressure favor the formation of the products.

Choose one answer

- ☐ $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
- ☐ $\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$
- ☐ $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- ☐ $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
- ☐ $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$

DK: Hvilket af følgende udsagn er korrekt for en 0.10 M opløsning af en stærk syre, HX?

EN: Which of the following statements is correct for a 0.10 M solution of a strong acid, HX?

Choose one answer

- ☐ $[X^-] \ll 0.10 \text{ M}$
- ☐ $[HX] > [H^+]$
- ☐ $\text{pH} = 1; [H^+] = 0.10 \text{ M}$
- ☐ $[HX] = 0.10 \text{ M}$
- ☐ $[H^+] \ll 0.10 \text{ M}$

DK: En opløsning fremstilles ved at blande 0.50 mol eddikesyre (CH_3COOH ; $K_a=1.80\times 10^{-5}$) med 0.500 mol natriumacetat (NaCH_3COO) og vand så det endelige volumen er 0.50 L. Til denne opløsning tilsættes 15.0 mL af en 1.00 M NaOH(aq) -opløsning. Hvad er pH i den endelige opløsning?

EN: A solution is prepared by mixing 0.50 mol acetic acid (CH_3COOH ; $K_a=1.80\times 10^{-5}$) with 0.500 mol sodium acetate (NaCH_3COO) and water such that the final volume is 0.50 L. To this solution 15.0 mL of a 1.00 M NaOH(aq) solution is added. What is the pH of the final solution?

Choose one answer

- ☐ 4.77
- ☐ 4.30
- ☐ 4.72
- ☐ 2.38
- ☐ 4.75

DK: K er ligevægtskonstanten for en kemisk reaktion. Ved hvilken temperatur findes $K=1$ når det vides om reaktionen at $\Delta H = 50 \text{ kJ/mol}$ og $\Delta S = 67 \text{ J/molK}$.

EN: K is the equilibrium constant for a chemical reaction. At which temperature is $K=1$ when it is known for the reaction that $\Delta H = 50 \text{ kJ/mol}$ and $\Delta S = 67 \text{ J/molK}$.

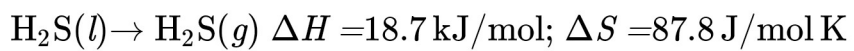
Choose one answer

- ☐ 273K
- ☐ 746 K
- ☐ 500 K
- ☐ 298 K
- ☐ 0 K

DK: Estimer kogepunktet for H_2S .



EN: Estimate the boiling point for H_2S .



Choose one answer

- ☐ 60°C
- ☐ -60°C
- ☐ 212°C
- ☐ 373°C
- ☐ -20°C

DK: Acetylen (C_2H_2) produceres industrielt blandt andet ved partiel forbrænding af methan (CH_4) efter (det ikke afstemte) reaktionsskema:



Hvis 100 L methan (STP) reageres med 150 L atmosfærisk luft (STP, d.v.s. $p(\text{O}_2)=0.2$ atm) hvor meget acetylen (STP) kan da fremstilles?

(STP: $t = 25^\circ\text{C}$ og $p = 1$ atm).

EN: Acetylene (C_2H_2) is produced industrially, for example by partial combustion of methane (CH_4) according to the (non-balanced) reaction scheme:



If 100 L (STP) of methane is reacted with 150 L of atmospheric air (STP, i.e. $p(\text{O}_2)=0.2$ atm), how much acetylene can be produced.

(STP: $t = 25^\circ\text{C}$ og $p = 1$ atm).

Choose one answer

- ☐ 50 L
- ☐ 30 L
- ☐ 100 L
- ☐ 10 L
- ☐ 20 L

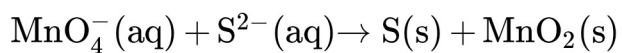
DK: Hvad er oxidationstrinnet for fosfor i forbindelserne A: PO; B: P₂O₃; C: P₂O₄; D: P₂O₅?

EN: What is the oxidation state of phosphorus in the compounds A: PO; B: P₂O₃; C: P₂O₄; D: P₂O₅?

Choose one answer

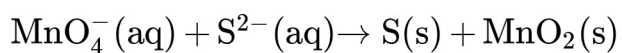
- ☐ A: +2; B: +4; C: +4; D: +5
- ☐ A: +1; B: +2; C: +3; D: +4
- ☐ A: +2; B: +3; C: +4; D: +5
- ☐ A: +3; B: +3; C: +4; D: +5
- ☐ A: 0; B: +1; C: +2; D: +3
- ☐ A: -1; B: +2; C: +3; D: +4
- ☐ A: +2; B: +3; C: +5; D: +5
- ☐ A: +2; B: +3; C: +4; D: 0
- ☐ A: 1; B: 1.5; C: 2; D: 2.5;

DK: Afstem følgende reaktion for oxidationen af sulfidioner til svovl med permanganat i basisk opløsning.



Hvad er summen af koefficienterne?
Brug de laveste mulige, heltallige koefficienter.

EN: Balance the following reaction for the oxidation sulfide ions with permanganate ions in basic solution:



What is the sum of the coefficients?
Use the smallest possible, integer coefficients.

Choose one answer

- ☐ 25
- ☐ 21
- ☐ 22
- ☐ 13
- ☐ 4

DK: Et galvanisk element konstrueres med en halvcelle bestående af $\text{Cr}^{3+}/\text{Cr}^{2+}$ ($E^0 = -0.41 \text{ V}$) og en halvcelle bestående af $\text{Cu}^{2+}/\text{Cu}^+$ ($E^0 = +0.15 \text{ V}$). Hvilket af følgende udsagn er korrekt?

EN: A galvanic element is constructed with a half-cell consisting of $\text{Cr}^{3+}/\text{Cr}^{2+}$ ($E^0 = -0.41 \text{ V}$) and a half-cell consisting of $\text{Cu}^{2+}/\text{Cu}^+$ ($E^0 = +0.15 \text{ V}$). Which of the following statements is true?

Choose one answer

- ☐ Cr^{2+} reduceres til Cr^{3+} . *Cr^{2+} is being reduced to Cr^{3+} .*
- ☐ Cu^{2+} oxideres til Cu^+ . *Cu^{2+} is being oxidized to Cu^+ .*
- ☐ Cr^{3+} oxideres til Cr^{2+} . *Cr^{3+} is being oxidized to Cr^{2+} .*
- ☐ Cu^{2+} reduceres til Cu^+ . *Cu^{2+} is being reduced to Cu^+ .*
- ☐ Elektroner strømmer fra Cu^{2+} to Cr^{3+} . *Electrons are flowing from the Cu^{2+} to Cr^{3+} .*

DK: Hvor mange isomere ketoner har molekylformlen $C_5H_{10}O$?

EN: *How many isomeric ketones have the molecular formula $C_5H_{10}O$?*

Choose one answer

☐ 2

☐ 1

☐ 3

☐ 4

☐ 5

☐ 6

